ConsensusScope: Safety Routing for Generative AI English Essay Feedback

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Abstract

Generative AI can reduce the workload of English essay feedback, but fluent suggestions may still change student meaning, add unsupported content, or overcorrect a draft. We present ConsensusScope, an interactive system for safety routing of AI-generated English essay feedback. The system treats each feedback item, rather than an essay score or final rewrite, as the unit of control: it normalizes candidates, extracts deploy-time safety signals, constructs an auditable Feedback Safety Graph, and applies a pilot-calibrated logistic release policy to route each item to auto_accept, teacher_review, or needs_more_evidence. To make this routing layer directly usable, we instantiate it as a Streamlit/FastAPI demo for single-essay review, batch review, teacher queueing, graph inspection, and report export. We evaluate routing with a stress suite, a public learner-corpus benchmark from JFLEG, CoNLL-2014, FCE, and W&I+LOCNESS, and a 30-item two-teacher pilot. Across 349,782 generated feedback candidates, the default policy auto-routes 32.9% while sending all constructed incorrect or unsafe candidates to review. The teacher pilot shows 0.857 auto-accept precision against teacher-safe items and 0.694 within-one-point teacher agreement. These results support graph-backed safety routing, not automatic essay scoring or teacher replacement.

1 Introduction

Generative AI is becoming a plausible assistant for English essay feedback, a task with a long history in grammatical error correction (GEC) benchmarks (Ng et al., 2014; Bryant et al., 2019), learner-writing corpora (Yannakoudakis et al., 2011; Napoles et al., 2017), and automated writing support (Madnani et al., 2018). The practical appeal is clear: teachers can obtain grammar, vocabulary, organization, and argument comments at essay scale. The unresolved bottleneck is safe release. A fluent AI comment can be wrong after a modal verb, shift the student's claim, add unsupported evidence, or give advice that is too broad to show directly to learners. Reviewing every suggestion removes the promised workload reduction; releasing every suggestion risks misleading students.

ConsensusScope formulates this bottleneck as item-level safety routing. Given an anonymized ESL draft and AI feedback candidates, the system routes each item to auto_accept, teacher_review, or needs_more_evidence. The decision unit is a concrete feedback item tied to a target span, local context, suggestion, rationale, and issue type, rather than an essay score or final rewrite.[1]

The core mechanism is Feedback Safety Graph Routing. ConsensusScope links the target span, AI suggestion, evidence status, active safety dimensions, and final route in an auditable graph. Deploy-time signals include meaning-preservation warnings, content-grounding risks, local-edit scope, tone and specificity, model agreement, and parse failures. The system therefore exposes why an item is released, reviewed, or flagged as needing more evidence.

We evaluate this middle layer with stress checks, a public learner-corpus benchmark over 349,782 generated feedback candidates, and a 30-item two-teacher diagnostic pilot. The evidence supports a narrow claim: ConsensusScope can auto-route low-risk feedback while keeping constructed unsafe items in the

[1]Code and hosted demo: https://github.com/yyting2006-ai/ConsensusScope and https://demo.consensusscope.cn/.

review queue. It does not replace teachers or measure classroom learning gains. The demo contributes a safety-routing formulation, a Feedback Safety Graph Routing algorithm with pilot-calibrated release policy, a usable Streamlit/FastAPI review system, and a reproducible evaluation package for teacher-supervised AI feedback.

2 Related Work

Learner writing and GEC. GEC benchmarks have provided standardized correction tasks and learner-writing data, including CoNLL-2014 (Ng et al., 2014), BEA-2019 W&I+LOCNESS (Bryant et al., 2019), FCE (Yannakoudakis et al., 2011), and JFLEG (Napoles et al., 2017). Earlier shared tasks and corpora such as HOO and NUCLE helped establish learner-text correction as a benchmarked NLP problem (Dale and Kilgarriff, 2011; Dahlmeier et al., 2013). Phrase-level edit extraction and scorer conventions such as M2 and ERRANT further standardized how original/corrected pairs are converted into edit records (Dahlmeier and Ng, 2012; Bryant et al., 2017). These datasets primarily evaluate correction quality. ConsensusScope uses their correction annotations only as offline evidence for a downstream routing question: whether risky or incorrect feedback candidates remain in a review queue.

Automated writing feedback. Automated writing systems can provide revision support, but classroom use depends on how learners and teachers interpret the feedback (Madnani et al., 2018; Ranalli, 2018). ConsensusScope follows this caution by avoiding a teacher-replacement claim and making AI feedback inspectable before it becomes student-facing.

LLM reliability and adjudication. Multi-output and judge-based methods can help select answers or evaluate generations (Wang et al., 2023; Zheng et al., 2023; Liu et al., 2023). However, consensus or judge confidence is not equivalent to educational safety. Our demo treats model agreement as one observable signal alongside issue type, evidence status, meaning-change warnings, unsupported-claim cues, tone cues, and parse errors.

Unlike GEC tools, essay scorers, or generic LLM-as-judge workflows, ConsensusScope routes each feedback item before it reaches a student and exposes graph evidence behind release or review decisions.

3 Method

The core method in ConsensusScope is the Feedback Safety Graph Router (FSGR), a three-module workflow for deciding whether an AI-generated writing-feedback item can be released to a student. FSGR addresses three connected deployment problems: heterogeneous model outputs, opaque safety evidence, and unreliable automatic release. Given an anonymized essay E and feedback candidates $\mathcal{C} = \{c_i\}_{i=1}^{n}$, Module 1 converts each raw candidate into a unified feedback item, Module 2 converts the item into graph-backed safety signals, and Module 3 converts those signals into a route.

Module 1: candidate normalization. The input to the first module is a raw feedback candidate c_i: one proposed comment or edit for a specific span in an ESL essay, produced by an LLM, a prompt variant, or a benchmark generator. Because raw candidates may omit rationales, use different issue labels, or expose different model metadata, FSGR normalizes each candidate into $f_i = (q_i, d_i, x_i, z_i, u_i, h_i, m_i, t_i, a_i)$. Here q_i is the feedback-item identifier, d_i is the essay identifier, x_i is the target span, z_i is the surrounding context, u_i is the AI suggestion, h_i is the rationale, m_i is the model source, t_i is the predicted issue type, and a_i is an optional agreement signal. The output of this module is therefore a comparable item f_i that can be inspected by the same downstream safety procedure regardless of which generator produced it.

Module 2: safety graph construction. The second module takes f_i as input and asks what observable evidence supports or blocks release. FSGR extracts a binary signal vector b_i for parse failure, evidence conflict, meaning-change cues, unsupported claims, whole-draft rewriting, tone warnings, teacher-dependent wording, broad target spans, and agreement risk. It then maps these signals to six graph dimensions: local edit, meaning preservation,

System type	Main object	Typical output	ConsensusScope distinction
GEC / revision tools	Sentence edit	Corrected text	Audits whether feedback should be student-facing
Essay scoring systems	Whole essay	Score or rubric label	Avoids grading and routes item-level AI feedback
LLM-as-judge workflows	Model output	Preference or quality score	Uses graph-backed deploy-time safety signals
Feedback dashboards	Review records	Tables or filters	Connects target span, suggestion, risk, route, and teacher action

Table 1: ConsensusScope is positioned as feedback safety routing rather than automatic correction, essay scoring, or generic model judging.

content grounding, pedagogical tone, feedback specificity, and model agreement. The module outputs an auditable graph G_i whose core path is target span → AI suggestion → active safety dimension → routing decision. Nodes store the span, context, suggestion, rationale, issue type, evidence signal, active dimensions, and route; edges store relations such as contains, is_revised_by, activates, and justifies_route.

Module 3: calibrated release routing. The third module takes the graph features from Module 2 and produces the final route $R(f_i) = (r_i, p_i, s_i, G_i)$. FSGR first computes a bounded risk score and a pilot-calibrated review probability:

$$s_i = \min(1, \sum_k \alpha_k b_{ik}), \quad p_i = \sigma(\beta_0 + \beta^\top x_i),$$

where x_i concatenates s_i with the active graph-dimension indicators and p_i is the calibrated probability that teacher review is needed, following the standard binary-response regression formulation (Cox, 1958). It then assigns the route:

$$r_i = \begin{cases} E, & \phi_i = 1, \\ V, & h(b_i) = 1 \vee p_i \geq \tau, \\ A, & \text{otherwise}, \end{cases}$$

where ϕ_i indicates parse or evidence-grounding failure, $h(\cdot)$ marks high-severity signals, and τ is selected on the two-teacher diagnostic pilot to preserve review-needed recall. The actions E, V, and A correspond to evidence request, teacher review, and auto release, respectively. The module outputs a risk level, recommended action, risk reasons, risk score, graph path, node/edge records, and a short explanation. Teacher ratings and public-corpus gold corrections are reserved for offline calibration and evaluation, not used as deploy-time labels.

4 Functionality

The Streamlit/FastAPI demo exposes the router as a teacher-facing workflow: single-essay review, batch review, AI feedback comparison, teacher queue, evaluation, and report export. Users inspect routed items, open graph paths, prioritize review queues, and save teacher decisions through the backend.

The intended users are English writing teachers, educational NLP developers, and researchers studying human-in-the-loop feedback. Teachers use the system to separate low-risk local edits from suggestions that still require professional judgment. Developers use normalized records to debug generative-feedback pipelines, and researchers can compare routing policies, risk signals, and teacher-aligned diagnostics.

The workflow is designed around teacher control rather than automatic release. In single-essay mode, a teacher can inspect how local spans, AI suggestions, and graph dimensions produce a route. In batch mode, the teacher queue prioritizes medium- and high-risk feedback items before student-facing reports are exported. This design makes the system useful even when the underlying feedback generator is replaced, because the routing layer only assumes a unified feedback schema.

Interfaces. The web interface is the primary demo surface for teachers. It exposes the essay view, feedback-item queue, graph explanation panel, batch upload, and student-report export. The FastAPI backend exposes the same routing functions for programmatic use, so developers can submit feedback items and retrieve the route, risk reasons, calibrated review probability, and graph record without using the Streamlit front end. The repository

also includes command-line scripts for regenerating public-corpus benchmarks and calibration artifacts.

5 End-to-End Use Case

Consider a teacher reviewing a comparative-literature essay written by an English learner. ConsensusScope receives the anonymized draft and a set of AI-generated feedback candidates: a safe local grammar edit, a vocabulary substitution that slightly changes the student's stance, an organization comment about paragraph order, and a content suggestion that introduces evidence not present in the essay. The use case is to decide, before anything is shown to the student, which items can be released directly and which items need teacher judgment.

ConsensusScope inserts a review-routing layer between generation and release. Each suggestion is converted to the unified feedback schema with its target span, local context, AI suggestion, rationale, issue type, and optional model agreement. The router then constructs a Feedback Safety Graph. A low-risk local grammar edit may activate the local-edit dimension and be routed to auto_accept. A vocabulary edit that changes the student's intended meaning activates meaning preservation and is routed to teacher_review. A content suggestion without support in the draft activates content grounding and is also routed to review. The teacher queue therefore focuses attention on items where professional judgment matters most.

The same workflow also supports batch review. When many essays are uploaded, the teacher can first inspect high-risk or evidence-poor feedback, then export a student-facing report containing only accepted or edited items. The exported audit record preserves the route, risk reasons, graph path, and teacher action, so later error analysis can identify which feedback types repeatedly require human intervention. This is the central application value of the system: it does not make AI feedback automatically correct, but it makes feedback release observable, reviewable, and safer for teacher-supervised use.

6 Evaluation

We evaluate ConsensusScope as a review-routing layer, not as a full feedback generator or a measure of student learning gains. The comparison question is whether Feedback Safety Graph Routing (FSGR) gives a better safety-efficiency trade-off than simpler policies such as review-all, low-risk release, or a more permissive release threshold. We measure unsafe-feedback recall in the review queue, auto-release share, and alignment with teacher-facing diagnostics.

Algorithmic stress checks. The packaged demo contains 15 feedback items with expected routes and 16 stress-test cases for dangerous suggestions such as thesis reversal, whole-essay rewriting, unsupported factual claims, harsh tone, low agreement, and parse failures. The router matches the expected action and risk labels on all 31 items; this is an algorithm sanity check, not a classroom study.

Public learner-corpus benchmark. We construct an offline routing benchmark from public learner-correction corpora: JFLEG (Napoles et al., 2017), CoNLL-2014 (Ng et al., 2014), FCE (Yannakoudakis et al., 2011), and W&I+LOCNESS from the BEA-2019 shared task (Bryant et al., 2019). For each original/corrected pair, we use edit-distance alignment (Levenshtein, 1966), following the edit-centric tradition of GEC evaluation (Dahlmeier and Ng, 2012; Bryant et al., 2017), to create feedback-level gold edits. From each gold edit, we generate one gold-derived correct candidate and constructed risky distractors: one meaning-change distractor for every edit and, for low-risk local GEC edits, one additional overcorrection distractor. Thus the 349,782 candidates are not newly collected student records or uncontrolled LLM generations; they are benchmark-derived feedback candidates built from 117,401 public gold edits plus 232,381 constructed risk distractors. The router sees only deploy-time feedback fields; gold labels are used afterward to check whether unsafe candidates were routed to review (Table 2).

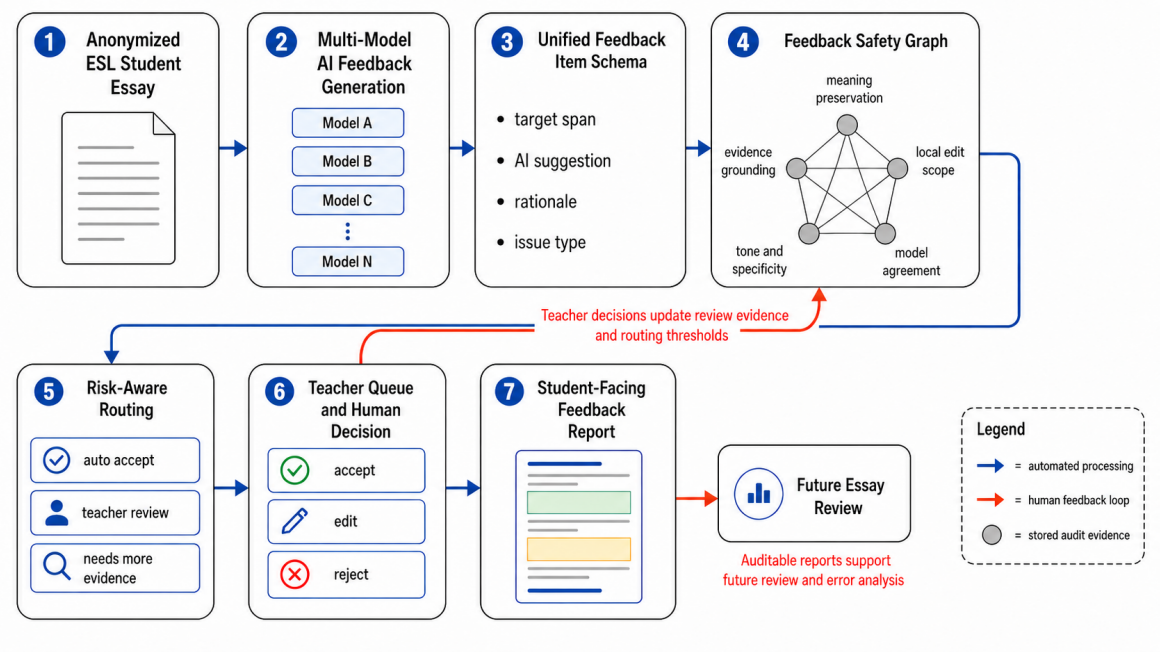

Figure 1: Feedback Safety Graph routing cycle in ConsensusScope. Blue arrows denote automated processing from anonymized essays and multi-model feedback to item-level routing decisions, while red arrows denote teacher feedback loops that update calibration evidence, review evidence, and future audit analysis.

Source	Records	Gold edits	Correct cand.	Risk dist.	Candidates	Auto	Review
JFLEG	1,501	3,938	3,938	7,649	11,587	0.320	0.680
CoNLL-2014	2,624	5,110	5,110	10,016	15,126	0.324	0.676
FCE	33,236	45,969	45,969	90,965	136,934	0.329	0.671
W&I+LOCNESS	38,692	62,384	62,384	123,751	186,135	0.329	0.671
Total	76,053	117,401	117,401	232,381	349,782	0.328	0.672

Table 2: Public learner-corpus routing benchmark. Candidate counts are derived from public gold edits: each edit contributes one correct candidate and one or two constructed risk distractors.

Two-teacher diagnostic pilot. To test whether the review routes align with human judgments, we collected a small blind pilot from two English teachers over 30 anonymized AI feedback items. Each teacher rated correctness, meaning preservation, student readiness, usefulness, clarity, and direct-release suitability on 1–5 scales. The ratings are offline diagnostics: individual teacher labels are not read at deploy time, while the aggregate pilot is used to fit the lightweight calibration artifact. The pilot exposed borderline auto-release patterns: teacher-dependent wording, semantic drift in vocabulary edits, and wrong local grammar corrections after modal verbs. On the 30-item pilot, the final router sends 76.7% of items to review, reaches 1.000 recall for teacher-marked review-needed and unsafe items, and obtains 0.857 auto-accept precision against teacher-safe items (Table 3). Within-one-point teacher agreement is 0.694.

Teacher-aligned diagnostic	Value
Feedback items	30
Rating rows	60
Auto share	0.233
Review share	0.767
Review-needed recall	1.000
Unsafe reviewed recall	1.000
Auto precision vs. teacher-safe	0.857
Within-one-point teacher agreement	0.694

Table 3: Two-teacher offline diagnostic pilot. Teacher ratings are used for offline calibration and diagnosis, not as per-item labels during deployment.

Case analysis. The pilot cases show the intended behavior: local phrase improvements can be auto-accepted, while thesis reversal, unsupported exam-score claims, and

Policy	Auto	Acc.	Review	Err. rev.
Accept low; review med/high	0.329	1.000	0.671	1.000
Accept low/med; review high	0.336	1.000	0.664	1.000
Review all	0.000	–	1.000	1.000

Table 4: Pooled review-routing utility over the public learner-corpus benchmark. Err. rev. is the share of observed incorrect or unsafe candidates routed to human review.

wrong modal-verb corrections are routed to review through meaning-preservation or content-grounding graph dimensions, without using teacher ratings at deploy time.

Review-routing utility. Table 4 compares three release policies over the same 349,782 feedback candidates. Review all is safe but gives no workload reduction; Accept low/med; review high is the more permissive baseline. The default FSGR policy accepts only low-risk local edits, auto-routes 32.9% of candidates, and preserves all constructed incorrect or unsafe candidates in the review queue. The 1.000 auto accuracy is a controlled diagnostic: correct candidates are derived from public gold corrections, and incorrect candidates are constructed risk distractors.

Interpretation. The evaluation should be read as review-routing evidence rather than a claim about fully automatic feedback quality. The public-corpus benchmark tests whether the router keeps constructed unsafe feedback out of automatic release; the teacher pilot checks whether the same signals align with human judgments on correctness, meaning preservation, readiness, and review necessity. The current policy is intentionally conservative: it sacrifices some automation rate in order to preserve all observed unsafe candidates in the review queue. This trade-off matches the target use case, where teacher time is limited but student-facing feedback should remain under human control.

7 Demo Scenario

The two-and-a-half-minute demo follows a teacher workflow: review one anonymized ESL draft, inspect graph paths, process a batch CSV, compare AI feedback, handle a high-risk item in Teacher Queue, and export a report.

8 Scope and Design Choices

ConsensusScope is deliberately designed as a routing and observability layer, not as a new essay scorer or a replacement for the feedback generator. This choice reflects the deployment setting: schools and developers may change the underlying LLM provider, prompt, or feedback generator, but they still need a stable mechanism for deciding which feedback items can be shown to students. The unified schema separates generation from review, while the graph record keeps the decision auditable.

The system also separates deploy-time signals from offline diagnostic labels. At deploy time, the router can use agreement, issue type, target-span scope, parse status, evidence availability, meaning-change cues, and tone/specificity signals. It cannot know whether an item is a false consensus or a minority correct case unless such labels are produced by later evaluation. This separation is important for honest reporting: aggregate teacher ratings may calibrate the release policy offline, but per-item teacher labels and public correction labels are not read when routing a new student-facing item.

These design choices define the contribution boundary. ConsensusScope does not claim that a particular LLM produces high-quality educational feedback. Instead, it shows how feedback from one or more generators can be converted into auditable routing decisions before reaching students. This makes the system suitable for classroom-facing deployments where teachers want the efficiency of generative feedback but still need a defensible review process.

9 Availability and Reproducibility

ConsensusScope is released under the MIT License. The repository, live demo, video, and API documentation are available at https://github.com/yyting2006-ai/ConsensusScope, https://demo.consensusscope.cn/, https://youtu.be/vp4Kj48bv8Q, and https://api.consensusscope.cn/docs. The hosted demo uses a self-hosted Streamlit/FastAPI deployment. The repository includes startup instructions and scripts/run_public_gec_benchmark.py; raw corpora and per-item derived outputs are not

committed.

10 Ethics and Limitations

ConsensusScope is a teacher-support tool, not an automatic essay scorer or teacher replacement. Users should upload only anonymized student writing. The benchmark validates graph-backed routing with public correction labels and constructed distractors; it does not measure learning gains, teacher satisfaction, time-on-task, or real LLM feedback quality. The 30-item two-teacher pilot is diagnostic, not a classroom study.

11 Conclusion

ConsensusScope demonstrates safety routing for generative AI English essay feedback: it converts feedback items into graph-backed release decisions so teachers can reduce review burden without giving up control over what students receive.

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A Supplementary Implementation Artifacts

This appendix is an artifact sheet for reproduction: pseudocode, calibrated weights, exported fields, and representative routed cases.

Algorithm A.1: Feedback Safety Graph Router for one feedback item

Input: normalized item f_i, local context z_i, optional rationale h_i, agreement signal a_i
Output: route r_i, review probability p_i, risk score s_i, graph G_i

1 Extract deploy-time signals b_i for parse failure, evidence conflict, meaning-change cues, unsupported claims, tone warnings, broad target spans, and agreement risk.
2 Map b_i to graph dimensions: local edit, meaning preservation, content grounding, pedagogical tone, feedback specificity, and model agreement.
3 Construct G_i with target-span, suggestion, evidence, dimension, and route nodes.
4 Compute $s_i = \min(1, \sum_k \alpha_k b_{ik})$.
5 Build feature vector $x_i = [s_i; \text{active dimensions}]$ and compute $p_i = \sigma(\beta_0 + \beta^\top x_i)$.
6 If parse or grounding failure is active, set $r_i = E$ (needs_more_evidence).
7 Else if a high-severity signal is active or $p_i \geq \tau$, set $r_i = V$ (teacher_review).
8 Else set $r_i = A$ (auto_accept).
9 Return r_i, p_i, s_i, risk reasons, graph path, and JSON node/edge records.

Figure A.1: Routing pseudocode used by the demo implementation. The actions E, V, and A denote evidence request, teacher review, and auto release.

Calibration feature	Weight
Intercept	-0.582
Risk score	0.861
Local-edit signal	-0.467
Meaning-risk signal	1.138
Grounding-risk signal	0.653
Specificity-risk signal	-0.047
Agreement-risk signal	0.000
Evidence-gap signal	0.000

Table A.1: Pilot-calibrated logistic release policy. The cutoff is 0.300 and is stored in routing_calibration.json.

Output field	Stored value
risk_level	Risk label
recommended_action	Release route
Calibrated review probability	Review probability
Active graph dimensions	Active dimensions
safety_graph_path	Explanation path
safety_graph_nodes	JSON node records
safety_graph_edges	JSON edge records

Table A.2: Core fields exported for each routed feedback item.

Case	Pattern	Graph dimensions	Route	Diagnostic use
FB-002	Local phrase improvement	Local edit	Auto-accept	Teachers rate safe
FB-003	Reverses student thesis	Meaning preservation	Review	Blocks meaning change
FB-005	Wrong modal correction	Local edit; meaning preservation	Review	Finds unsafe local edit
FB-008	Unsupported exam-score claim	Meaning preservation; content grounding	Review	Blocks unsupported evidence

Table A.3: Representative pilot cases. Teacher ratings are offline diagnostics and are not per-item inputs during deploy-time routing.